

DAG Pipeline Declared by `pipeline.yaml` |

Single-project pipelines read `infrastructure/core/pipeline/pipeline.yaml`. `scripts/runner/execute_pipeline.py` expands the declarative DAG, applies tag filters (`--core-only` skips llm stages), checkpoints between nodes, then dispatches numbered scripts (`scripts/NN_*.py`) or builtin methods (`_run_clean_outputs`).

The **default YAML graph contains ten named stages** (plus telemetry configuration metadata):

1. **Clean Output Directories** — wipes prior `projects/<name>/output/` + delivered `output/<name>/` paths so stale PDFs cannot satisfy validation.
2. **Environment Setup** (`00_setup_environment.py`) — Python/uv probing, toolchain discovery, scaffolding directories, `PYTHONPATH` wiring.
3. **Infrastructure Tests** (`01_run_tests.py --infra-only`) — tests/ suite with the configured 60% source-coverage floor.

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4. **Project Tests** (`01_run_tests.py --project-only`) — isolated per-project suites with each project's declared floor (90% for this exemplar).
5. **Project Analysis** (`02_run_analysis.py`) — lexicographically ordered `projects/<name>/scripts/*.py`, each a thin orchestrator (`src/` does real work).
6. **PDF Rendering** (`03_render_pdf.py`) — Pandoc → XeLaTeX loop, bibliography assembly, injected variables from Stage 02 artefacts.
7. **Output Validation** (`04_validate_output.py`) — PDF structure, manifests, Markdown hygiene.
8. **LLM Scientific Review** (`06_llm_review.py --reviews-only; tags: llm`) — executive + quality critiques via local Ollama; `allow_skip: true`.
9. **LLM Translations** (`06_llm_review.py --translations-only; tags llm`, same dependency edges) — multilingual abstract expansion.

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10. **Copy Outputs** (`05_copy_outputs.py`) — reproducible snapshots into canonical `output/<project>/`.

Two LLM nodes intentionally share one script module with orthogonal CLI switches; both depend only on validation so they can parallelize logically while remaining optional.

Executive reporting (`scripts/pipeline/stage_07_executive_report.py`) is **not** a YAML node inside the single-project executor. `--all-projects / execute_multi_project.py` invokes it once after iterating projects, consolidating cross-project KPIs dashboards.

Topological order therefore differs slightly from lexical script numbering (e.g., `copy` executes after validation even though script 05 precedes 06 lexically).

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Stage Highlights

Infrastructure vs project tests. Splitting pytest invocations isolates flaky infra regressions (`MAX_TEST_FAILURES` knobs) from zero-tolerance gates on domain code (`max_project_test_failures` default 0 declared in YAML front-matter/testing blocks).

Stage 02 illustration. The analysis stage is deliberately concrete rather than a hypothetical diagram factory: each canonical project ships real behaviour at this node. `template_autoresearch_project` runs readiness validation; `template_search_project` merges remote literature JSON, generates scripted figures (`y_generate_search_figures.py`), and writes manifests; `template_code_project` emits optimization plots; and `template_prose_project` triggers structural validation scaffolding. The pipeline shape is identical across all four—only the Stage 02 payload differs—which is exactly what lets one orchestrator serve heterogeneous research domains.

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Interactive Orchestration

run.sh

Thin wrapper invoking `python -m infrastructure.orchestration`. Offers:

- ▶ per-project staged execution,
- ▶ chained digits (234 shorthand),
- ▶ multi-project grid (a-d presets),
- ▶ graceful quit / resume parity with `scripts/README.md`.

Selecting **d alone** after a passing multi-project run exits immediately once summaries print—avoiding repetitive menu redraw.

secure_run.sh

Executes Python secure path: standard pipeline artefact reproduction **then** invokes `run_secure_pipeline` for steganographic PDF hardening (`infrastructure.steganography`). Original PDFs stay immutable; hardened companions carry QR overlays plus hash manifests sidecars.